

Computational Geophysics: What is the best strategy for my problem?

Short Course

Heiner Igel

September 20, 2026

Department of Earth and Environmental Sciences
Ludwig-Maximilians-University Munich

What I would like to ask you ... (please raise your hand!)

- Who would be able to write down a simple wave or advection equation?
- Who had a course on numerical analysis?
- Who has programmed in Python?
- Who has NEVER used a Jupyter Notebook?
- Who uses a community code for simulations (e.g., specfem, or other) ?
- Who has ever run a parallel program (cpu or gpu)?
- Who has seen the movie "The Odyssey"?

What I would like to ask you ... (please raise your hand!)

- Who would be able to write down a simple wave or advection equation?
- Who had a course on numerical analysis?
- Who has programmed in Python?
- Who has NEVER used a Jupyter Notebook?
- Who uses a community code for simulations (e.g., specfem, or other) ?
- Who has ever run a parallel program (cpu or gpu)?
- Who has seen the movie "The Odyssey"?

What I would like to ask you ... (please raise your hand!)

- Who would be able to write down a simple wave or advection equation?
- Who had a course on numerical analysis?
- Who has programmed in Python?
- Who has NEVER used a Jupyter Notebook?
- Who uses a community code for simulations (e.g., specfem, or other) ?
- Who has ever run a parallel program (cpu or gpu)?
- Who has seen the movie "The Odyssey"?

What I would like to ask you ... (please raise your hand!)

- Who would be able to write down a simple wave or advection equation?
- Who had a course on numerical analysis?
- Who has programmed in Python?
- Who has NEVER used a Jupyter Notebook?
- Who uses a community code for simulations (e.g., specfem, or other) ?
- Who has ever run a parallel program (cpu or gpu)?
- Who has seen the movie "The Odyssey"?

What I would like to ask you ... (please raise your hand!)

- Who would be able to write down a simple wave or advection equation?
- Who had a course on numerical analysis?
- Who has programmed in Python?
- Who has NEVER used a Jupyter Notebook?
- Who uses a community code for simulations (e.g., specfem, or other) ?
- Who has ever run a parallel program (cpu or gpu)?
- Who has seen the movie "The Odyssey"?

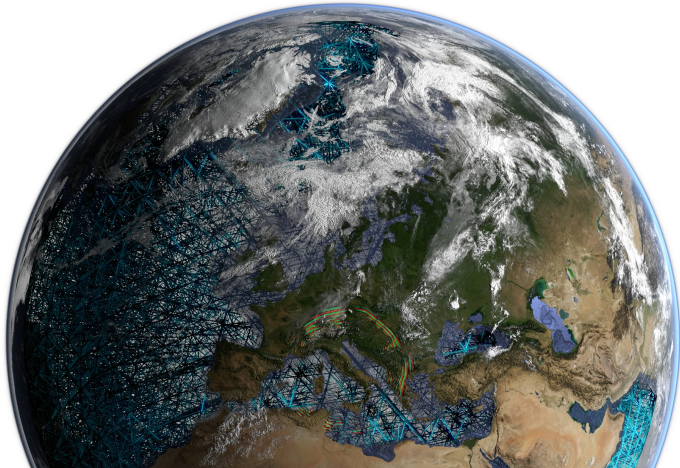
What I would like to ask you ... (please raise your hand!)

- Who would be able to write down a simple wave or advection equation?
- Who had a course on numerical analysis?
- Who has programmed in Python?
- Who has NEVER used a Jupyter Notebook?
- Who uses a community code for simulations (e.g., specfem, or other) ?
- Who has ever run a parallel program (cpu or gpu)?
- Who has seen the movie "The Odyssey"?

What I would like to ask you ... (please raise your hand!)

- Who would be able to write down a simple wave or advection equation?
- Who had a course on numerical analysis?
- Who has programmed in Python?
- Who has NEVER used a Jupyter Notebook?
- Who uses a community code for simulations (e.g., specfem, or other) ?
- Who has ever run a parallel program (cpu or gpu)?
- Who has seen the movie "The Odyssey"?

Introduction



Course Content

- Part I
 - Introduction - **Motivation**
 - **Fundamentals** of wave propagation (**wave equations, analytical solutions, reciprocity, superposition principle, dispersion, homogenization**)
- Part II
 - The **finite-difference** method
 - The **pseudospectral** method
 - Linear **finite-element** method
- Part III
 - The **spectral-element** method
 - The **finite-volume** method
 - The **discontinuous Galerkin** method
 - Outlook

Timing Today (approximate)

- 9:00 - 10:30 - Part I
- 10:30 - 11:00 Break
- 11:00 - 12:00 - Part II - A
- 12:00 - 14:00 Lunch Break
- 14:00 - 15:30 Part II - B
- 15:30 - 16:00 Break
- 16:00 - $\approx$ 17:00 Part III
- 17:00+ Optional discussions



What you get ...

- Basic ideas of **discretization**, **size** of a problem, designing a solution
- The fundamentals of **wave propagation** (what should I see in a wavefield?)
- What is in the **engine** of simulation algorithms?
- What are the major parameters, where are **dangers** and **traps**?
- What is a good strategy **to know my simulation is right**?
- What is the difference between the various approaches, **pros and cons**?
- What are the **simplest programs** to see how things work?

What you get ...

- Basic ideas of **discretization**, **size** of a problem, designing a solution
- The fundamentals of **wave propagation** (what should I see in a wavefield?)
- What is in the **engine** of simulation algorithms?
- What are the major parameters, where are **dangers** and **traps**?
- What is a good strategy **to know my simulation is right**?
- What is the difference between the various approaches, **pros and cons**?
- What are the **simplest programs** to see how things work?

What you get ...

- Basic ideas of **discretization**, **size** of a problem, designing a solution
- The fundamentals of **wave propagation** (what should I see in a wavefield?)
- What is in the **engine** of simulation algorithms?
- What are the major parameters, where are **dangers** and **traps**?
- What is a good strategy **to know my simulation is right**?
- What is the difference between the various approaches, **pros and cons**?
- What are the **simplest programs** to see how things work?

What you get ...

- Basic ideas of **discretization**, **size** of a problem, designing a solution
- The fundamentals of **wave propagation** (what should I see in a wavefield?)
- What is in the **engine** of simulation algorithms?
- What are the major parameters, where are **dangers** and **traps**?
- What is a good strategy **to know my simulation is right**?
- What is the difference between the various approaches, **pros and cons**?
- What are the **simplest programs** to see how things work?

What you get ...

- Basic ideas of **discretization**, **size** of a problem, designing a solution
- The fundamentals of **wave propagation** (what should I see in a wavefield?)
- What is in the **engine** of simulation algorithms?
- What are the major parameters, where are **dangers** and **traps**?
- What is a good strategy **to know my simulation is right**?
- What is the difference between the various approaches, **pros and cons**?
- What are the **simplest programs** to see how things work?

What you get ...

- Basic ideas of **discretization**, **size** of a problem, designing a solution
- The fundamentals of **wave propagation** (what should I see in a wavefield?)
- What is in the **engine** of simulation algorithms?
- What are the major parameters, where are **dangers** and **traps**?
- What is a good strategy **to know my simulation is right**?
- What is the difference between the various approaches, **pros and cons**?
- What are the **simplest programs** to see how things work?

What you get ...

- Basic ideas of **discretization**, **size** of a problem, designing a solution
- The fundamentals of **wave propagation** (what should I see in a wavefield?)
- What is in the **engine** of simulation algorithms?
- What are the major parameters, where are **dangers** and **traps**?
- What is a good strategy **to know my simulation is right**?
- What is the difference between the various approaches, **pros and cons**?
- What are the **simplest programs** to see how things work?

What you do not get ...

- How specific **community solvers** work (e.g., specfem, seissol, salvus, etc.)
- **Classic approaches** like ray tracing or reflectivity methods
- **Codes** for 3D solutions

What you do not get ...

- How specific **community solvers** work (e.g., specfem, seissol, salvus, etc.)
- **Classic approaches** like ray tracing or reflectivity methods
- **Codes** for 3D solutions

What you do not get ...

- How specific **community solvers** work (e.g., specfem, seissol, salvus, etc.)
- **Classic approaches** like ray tracing or reflectivity methods
- **Codes** for 3D solutions

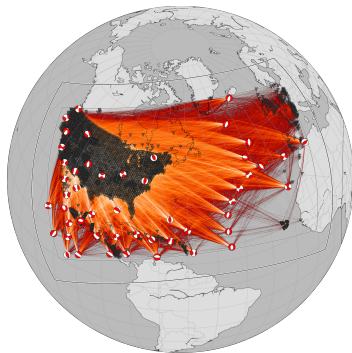
What is Computational Seismology?

We define **computational seismology** such that it **involves the complete solution of the seismic wave propagation (and rupture) problem for arbitrary 3-D models by numerical means.**

What is not covered ... but you can do tomography with ...

- **Ray-theoretical** methods
- **Quasi-analytical** methods (e.g., normal modes, reflectivity method)
- **Frequency-domain** solutions
- **Boundary integral** equation methods
- **Discrete particle** methods

These methods are important for **benchmarking** numerical solutions!



Who needs Computational Seismology

Many problems rely on the analysis of **elastic wavefields**

- **Global seismology** and tomography of the Earth's interior
- The quantification of **strong ground motion - seismic hazard**
- The understanding of the **earthquake source process**
- The monitoring of **volcanic processes** and the forecasting of eruptions
- **Earthquake early warning** systems
- **Tsunami early warning** systems
- Local, regional, and global **earthquake services**
- Global monitoring of **nuclear tests**
- **Laboratory scale analysis** of seismic events

Who needs Computational Seismology

Many problems rely on the analysis of **elastic wavefields**

- **Global seismology** and tomography of the Earth's interior
- The quantification of **strong ground motion - seismic hazard**
- The understanding of the **earthquake source process**
- The monitoring of **volcanic processes** and the forecasting of eruptions
- **Earthquake early warning** systems
- **Tsunami early warning** systems
- Local, regional, and global **earthquake services**
- Global monitoring of **nuclear tests**
- **Laboratory scale analysis** of seismic events

Who needs Computational Seismology

Many problems rely on the analysis of **elastic wavefields**

- **Global seismology** and tomography of the Earth's interior
- The quantification of **strong ground motion - seismic hazard**
- The understanding of the **earthquake source process**
- The monitoring of **volcanic processes** and the forecasting of eruptions
- **Earthquake early warning** systems
- **Tsunami early warning** systems
- Local, regional, and global **earthquake services**
- Global monitoring of **nuclear tests**
- **Laboratory scale analysis** of seismic events

Who needs Computational Seismology

Many problems rely on the analysis of **elastic wavefields**

- **Global seismology** and tomography of the Earth's interior
- The quantification of **strong ground motion - seismic hazard**
- The understanding of the **earthquake source process**
- The monitoring of **volcanic processes** and the forecasting of eruptions
- **Earthquake early warning** systems
- **Tsunami early warning** systems
- Local, regional, and global **earthquake services**
- Global monitoring of **nuclear tests**
- **Laboratory scale analysis** of seismic events

Who needs Computational Seismology

Many problems rely on the analysis of **elastic wavefields**

- **Global seismology** and tomography of the Earth's interior
- The quantification of **strong ground motion - seismic hazard**
- The understanding of the **earthquake source process**
- The monitoring of **volcanic processes** and the forecasting of eruptions
- **Earthquake early warning** systems
- **Tsunami early warning** systems
- Local, regional, and global **earthquake services**
- Global monitoring of **nuclear tests**
- **Laboratory scale analysis** of seismic events

Who needs Computational Seismology

Many problems rely on the analysis of **elastic wavefields**

- **Global seismology** and tomography of the Earth's interior
- The quantification of **strong ground motion - seismic hazard**
- The understanding of the **earthquake source process**
- The monitoring of **volcanic processes** and the forecasting of eruptions
- **Earthquake early warning** systems
- **Tsunami early warning** systems
- Local, regional, and global **earthquake services**
- Global monitoring of **nuclear tests**
- **Laboratory scale analysis** of seismic events

Who needs Computational Seismology

Many problems rely on the analysis of **elastic wavefields**

- **Global seismology** and tomography of the Earth's interior
- The quantification of **strong ground motion - seismic hazard**
- The understanding of the **earthquake source process**
- The monitoring of **volcanic processes** and the forecasting of eruptions
- **Earthquake early warning** systems
- **Tsunami early warning** systems
- Local, regional, and global **earthquake services**
- Global monitoring of **nuclear tests**
- **Laboratory scale analysis** of seismic events

Who needs Computational Seismology

Many problems rely on the analysis of **elastic wavefields**

- **Global seismology** and tomography of the Earth's interior
- The quantification of **strong ground motion - seismic hazard**
- The understanding of the **earthquake source process**
- The monitoring of **volcanic processes** and the forecasting of eruptions
- **Earthquake early warning** systems
- **Tsunami early warning** systems
- Local, regional, and global **earthquake services**
- Global monitoring of **nuclear tests**
- **Laboratory scale analysis** of seismic events

Who needs Computational Seismology

Many problems rely on the analysis of **elastic wavefields**

- **Global seismology** and tomography of the Earth's interior
- The quantification of **strong ground motion - seismic hazard**
- The understanding of the **earthquake source process**
- The monitoring of **volcanic processes** and the forecasting of eruptions
- **Earthquake early warning** systems
- **Tsunami early warning** systems
- Local, regional, and global **earthquake services**
- Global monitoring of **nuclear tests**
- **Laboratory scale analysis** of seismic events

Who needs Computational Seismology (cont'd)

(...)

- Ocean generated **noise measurements** and cross-correlation techniques
- **Environmental** seismology
- **Planetary seismology** - Apollo, INSIGHT
- **Exploration geophysics**, reservoir scale seismics
- **Geotechnical engineering** (non-destructive testing, small scale tomography)
- **Medical applications**, breast cancer detection, reverse acoustics

Who needs Computational Seismology (cont'd)

(...)

- Ocean generated **noise measurements** and cross-correlation techniques
- **Environmental** seismology
- Planetary seismology - Apollo, INSIGHT
- Exploration geophysics, reservoir scale seismics
- Geotechnical engineering (non-destructive testing, small scale tomography)
- Medical applications, breast cancer detection, reverse acoustics

Who needs Computational Seismology (cont'd)

(...)

- Ocean generated **noise measurements** and cross-correlation techniques
- **Environmental** seismology
- **Planetary seismology - Apollo, INSIGHT**
- Exploration geophysics, reservoir scale seismics
- Geotechnical engineering (non-destructive testing, small scale tomography)
- Medical applications, breast cancer detection, reverse acoustics

Who needs Computational Seismology (cont'd)

(...)

- Ocean generated **noise measurements** and cross-correlation techniques
- **Environmental** seismology
- **Planetary seismology - Apollo, INSIGHT**
- **Exploration geophysics**, reservoir scale seismics
- **Geotechnical engineering** (non-destructive testing, small scale tomography)
- **Medical applications**, breast cancer detection, reverse acoustics

Who needs Computational Seismology (cont'd)

(...)

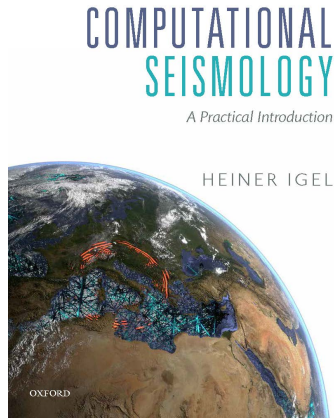
- Ocean generated **noise measurements** and cross-correlation techniques
- **Environmental** seismology
- **Planetary seismology - Apollo, INSIGHT**
- **Exploration geophysics**, reservoir scale seismics
- **Geotechnical engineering** (non-destructive testing, small scale tomography)
- **Medical applications**, breast cancer detection, reverse acoustics

Who needs Computational Seismology (cont'd)

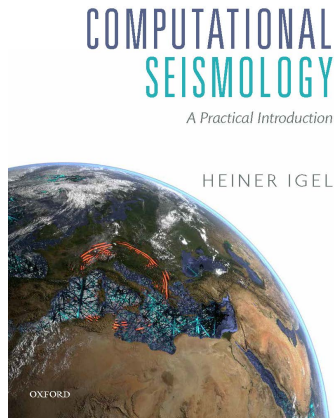
(...)

- Ocean generated **noise measurements** and cross-correlation techniques
- **Environmental** seismology
- **Planetary seismology** - **Apollo, INSIGHT**
- **Exploration geophysics**, reservoir scale seismics
- **Geotechnical engineering** (non-destructive testing, small scale tomography)
- **Medical applications**, breast cancer detection, reverse acoustics

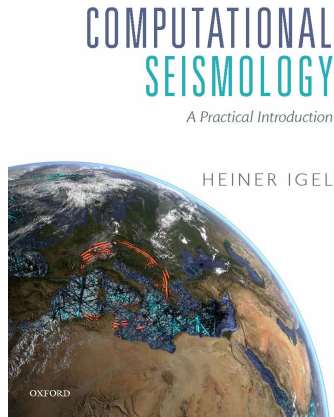
- Igel, **Computational Seismology: A Practical Introduction** (Oxford University Press, 2016, 2nd edition 2028)
- Shearer, **Introduction to Seismology** (3rd edition, 2019)
- Aki and Richards, **Quantitative Seismology** (1st edition, 1980)
- Mozco, **The Finite-Difference Modelling of Earthquake Motions** (Cambridge University Press)
- Fichtner, **Full Seismic Waveform Modelling and Inversion** (Springer Verlag, 2010).



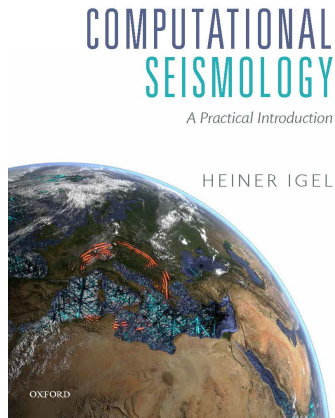
- Igel, **Computational Seismology: A Practical Introduction** (Oxford University Press, 2016, 2nd edition 2028)
- Shearer, **Introduction to Seismology** (3rd edition, 2019)
- Aki and Richards, **Quantitative Seismology** (1st edition, 1980)
- Mozco, **The Finite-Difference Modelling of Earthquake Motions** (Cambridge University Press)
- Fichtner, **Full Seismic Waveform Modelling and Inversion** (Springer Verlag, 2010).



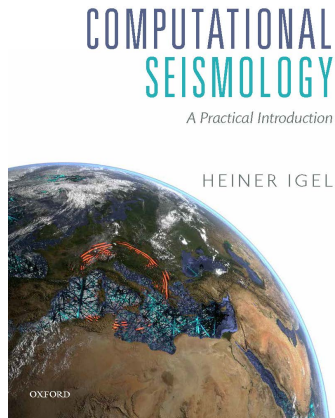
- Igel, **Computational Seismology: A Practical Introduction** (Oxford University Press, 2016, 2nd edition 2028)
- Shearer, **Introduction to Seismology** (3rd edition, 2019)
- Aki and Richards, **Quantitative Seismology** (1st edition, 1980)
- Mozco, **The Finite-Difference Modelling of Earthquake Motions** (Cambridge University Press)
- Fichtner, **Full Seismic Waveform Modelling and Inversion** (Springer Verlag, 2010).



- Igel, **Computational Seismology: A Practical Introduction** (Oxford University Press, 2016, 2nd edition 2028)
- Shearer, **Introduction to Seismology** (3rd edition, 2019)
- Aki and Richards, **Quantitative Seismology** (1st edition, 1980)
- Mozco, **The Finite-Difference Modelling of Earthquake Motions** (Cambridge University Press)
- Fichtner, **Full Seismic Waveform Modelling and Inversion** (Springer Verlag, 2010).



- Igel, **Computational Seismology: A Practical Introduction** (Oxford University Press, 2016, 2nd edition 2028)
- Shearer, **Introduction to Seismology** (3rd edition, 2019)
- Aki and Richards, **Quantitative Seismology** (1st edition, 1980)
- Mozco, **The Finite-Difference Modelling of Earthquake Motions** (Cambridge University Press)
- Fichtner, **Full Seismic Waveform Modelling and Inversion** (Springer Verlag, 2010).





Computers, Waves, Simulations: A Practical Introduction to Numerical Methods using Python



Instructor: [Heiner Igel](#)

Enroll now

27,562 already enrolled

9-week course in
Computational wave
propagation on
COURSERA. Covers the
finite-difference,
pseudospectral,
finite-element, and
spectral-element method.

Why numerical methods?

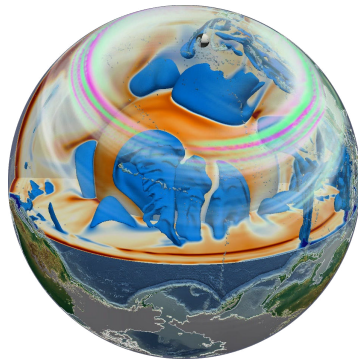


Photo ©Überraschungsbilder via Wikimedia commons

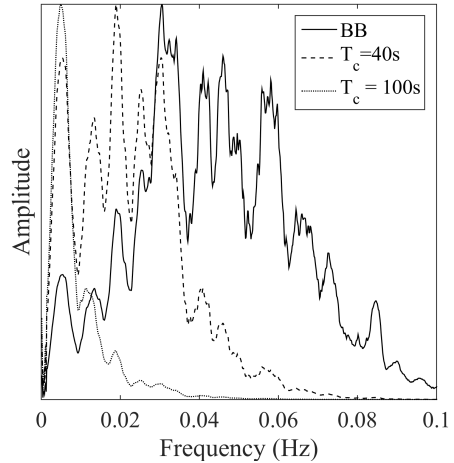
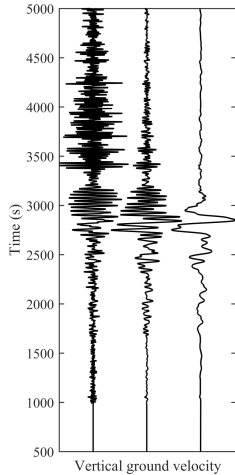
Computational Seismology, Memory, and Compute Power

Numerical solutions necessitate the **discretization** of Earth models. Estimate how much **memory** is required to store the Earth model and the required displacement fields.

Are we talking laptop or supercomputer?

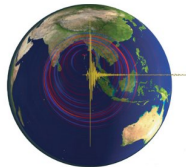


Seismic Wavefield Observations



Exercise: Sampling a global seismic wavefield

- The highest frequencies that we observe for global wave fields is 1Hz.
- We assume a homogeneous Earth (radius 6371km).
- P velocity $v_p = 10\text{km/s}$ and the v_p/v_s ratio is $\sqrt{3}$
- We want to use 20 **grid points (cells) per wavelength**
- How many grid cells would you need (assume cubic cells).
- What would be their size?
- How much memory would you need to store one such field (e.g., density in single precision).



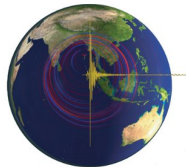
You may want to make use of

$$c = \frac{\lambda}{T} = \lambda f = \frac{\omega}{k}$$



Exercise: Sampling a global seismic wavefield

- The highest frequencies that we observe for global wave fields is 1Hz.
- We assume a homogeneous Earth (radius 6371km).
- P velocity $v_p = 10\text{km/s}$ and the v_p/v_s ratio is $\sqrt{3}$
- We want to use 20 **grid points (cells) per wavelength**
- How many grid cells would you need (assume cubic cells).
- What would be their size?
- How much memory would you need to store one such field (e.g., density in single precision).



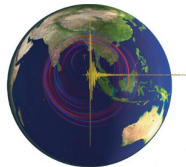
You may want to make use of

$$c = \frac{\lambda}{T} = \lambda f = \frac{\omega}{k}$$



Exercise: Sampling a global seismic wavefield

- The highest frequencies that we observe for global wave fields is 1Hz.
- We assume a homogeneous Earth (radius 6371km).
- P velocity $v_p = 10\text{km/s}$ and the v_p/v_s ratio is $\sqrt{3}$
- We want to use 20 **grid points (cells) per wavelength**
- How many grid cells would you need (assume cubic cells).
- What would be their size?
- How much memory would you need to store one such field (e.g., density in single precision).



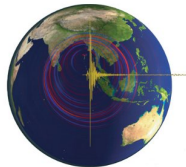
You may want to make use of

$$c = \frac{\lambda}{T} = \lambda f = \frac{\omega}{k}$$



Exercise: Sampling a global seismic wavefield

- The highest frequencies that we observe for global wave fields is 1Hz.
- We assume a homogeneous Earth (radius 6371km).
- P velocity $v_p = 10\text{km/s}$ and the v_p/v_s ratio is $\sqrt{3}$
- We want to use 20 **grid points (cells) per wavelength**
- How many grid cells would you need (assume cubic cells).
- What would be their size?
- How much memory would you need to store one such field (e.g., density in single precision).



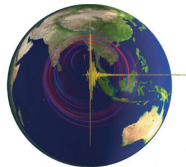
You may want to make use of

$$c = \frac{\lambda}{T} = \lambda f = \frac{\omega}{k}$$



Exercise: Sampling a global seismic wavefield

- The highest frequencies that we observe for global wave fields is 1Hz.
- We assume a homogeneous Earth (radius 6371km).
- P velocity $v_p = 10\text{km/s}$ and the v_p/v_s ratio is $\sqrt{3}$
- We want to use 20 **grid points (cells) per wavelength**
- How many grid cells would you need (assume cubic cells).
- What would be their size?
- How much memory would you need to store one such field (e.g., density in single precision).



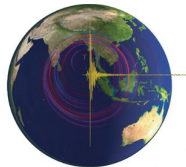
You may want to make use of

$$c = \frac{\lambda}{T} = \lambda f = \frac{\omega}{k}$$



Exercise: Sampling a global seismic wavefield

- The highest frequencies that we observe for global wave fields is 1Hz.
- We assume a homogeneous Earth (radius 6371km).
- P velocity $v_p = 10\text{km/s}$ and the v_p/v_s ratio is $\sqrt{3}$
- We want to use 20 **grid points (cells) per wavelength**
- How many grid cells would you need (assume cubic cells).
- What would be their size?
- How much memory would you need to store one such field (e.g., density in single precision).



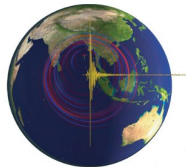
You may want to make use of

$$c = \frac{\lambda}{T} = \lambda f = \frac{\omega}{k}$$



Exercise: Sampling a global seismic wavefield

- The highest frequencies that we observe for global wave fields is 1Hz.
- We assume a homogeneous Earth (radius 6371km).
- P velocity $v_p = 10\text{km/s}$ and the v_p/v_s ratio is $\sqrt{3}$
- We want to use 20 **grid points (cells) per wavelength**
- How many grid cells would you need (assume cubic cells).
- What would be their size?
- How much memory would you need to store one such field (e.g., density in single precision).



You may want to make use of

$$c = \frac{\lambda}{T} = \lambda f = \frac{\omega}{k}$$



Exercise:

```
# Physics

# Earth's Velocity (S waves have smaller wavelength)
c = 10./np.sqrt(3) # km/s

# Target Frequency
T = 1. # Hz (1 Hz is usually the highets frequency of global wave fields)

# Wavelength
lam = c * T # in km

print(' Wavelength: ', lam, ' km ')
```

Wavelength: 5.773502691896258 km

```
# Computational method
npts = 20 # number of grid points per wavelength

# Required spatial discretization
dx = lam/npts # in km

# Size of a volume cell
dv = dx**3 # in km**3

print(' Size of Volume Cell',dv, 'km**3 ')
```

Size of Volume Cell 0.024056261216234418 km**3

```
# Memory requirement (Volume Earth / dv * 8 bytes)
mem = 4./3.*np.pi*6371**3/dv*8

print(' Memory requirement : ', mem/1000/1000/1000/1000, 'TBytes')
```

Memory requirement : 360.22452769668104 TBytes

Take a few minutes **with your neighbor** to think of your current or future simulation problem (whatever physical process). What is the **physical scale** of the problem? **2D or 3D**? What are the **wavelengths** involved? How much **memory** per field (elastic parameters, displacement)? **Laptop or supercomputer**?

Exercise: Solution

```
# Physics

# Earth's Velocity (S waves have smaller wavelength)
c = 10./np.sqrt(3) # km/s

# Target Frequency
T = 1. # Hz (1 Hz is usually the highets frequency of global wave fields)

# Wavelength
lam = c * T # in km

print(' Wavelength: ', lam, ' km ')
```

Wavelength: 5.773502691896258 km

```
# Computational method
npts = 20 # number of grid points per wavelength

# Required spatial discretization
dx = lam/npts # in km

# Size of a volume cell
dv = dx**3 # in km**3

print(' Size of Volume Cell',dv, 'km**3 ')
```

Size of Volume Cell 0.024056261216234418 km**3

```
# Memory requirement (Volume Earth / dv * 8 bytes)
mem = 4./3.*np.pi*6371**3/dv*8

print(' Memory requirement : ', mem/1000/1000/1000/1000, 'TBytes')
```

Memory requirement : 360.22452769668104 TBytes

Results (@ $T = 1s$) : 360
TBytes

Results (@ $T = 10s$) : 360
GBytes

Results (@ $T = 100s$) : 360
MBytes

Computational Seismology, Memory, and Compute Power



High-Performance Computing Milestones (1960–2019)

Floating point
operations
per second

exaFLOP
 1×10^{18}

petaFLOP
 1×10^{15}

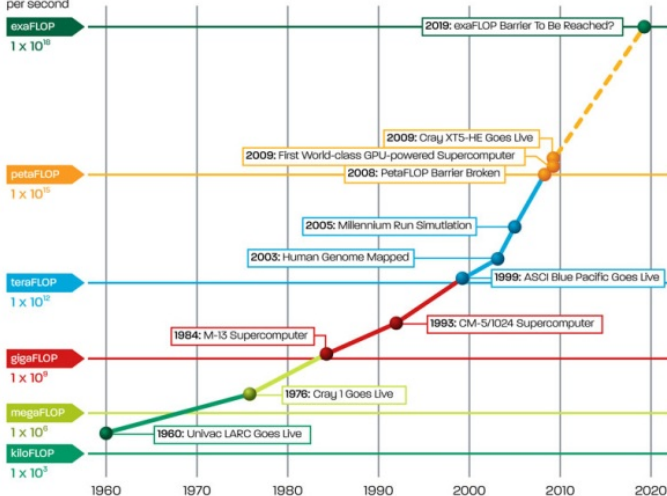
teraFLOP
 1×10^{12}

gigaFLOP
 1×10^9

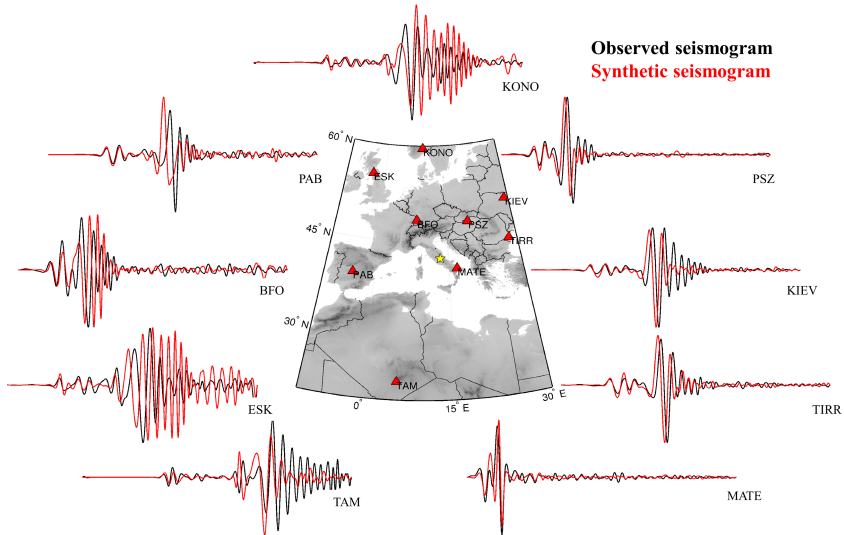
megaFLOP
 1×10^6

kiloFLOP
 1×10^3

1960: 1 MFlops
1970: 10MFlops
1980: 100MFlops
1990: 1 GFlops
1998: 1 TFlops
2008: 1 Pflops
2021: 1 EFlops



The Ultimate Goal: Matching Wavefield Observations



A Bit of Wave Physics

Acoustic wave equation: no source

Acoustic wave equation

$$\partial_t^2 p = c^2 \Delta p + s$$

$p \rightarrow p(\mathbf{x}, t)$, pressure

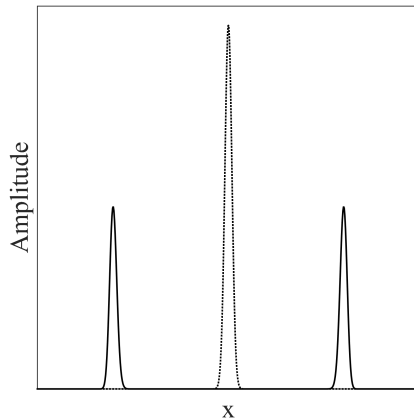
$c \rightarrow c(\mathbf{x})$, velocity

$s \rightarrow s(\mathbf{x}, t)$, source term

Initial conditions

$$p(\mathbf{x}, t = 0) = p_0(\mathbf{x}, t)$$

$$\partial_t p(\mathbf{x}, t = 0) = 0$$

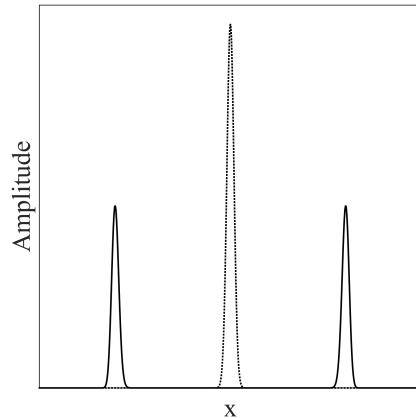


Snapshot of $p(\mathbf{x}, t)$ (solid line) after some time for initial condition $p_0(\mathbf{x}, t)$ (Gaussian, dashed line), 1D case.

Acoustic wave equation: no source

Solution

$$p(x, t) = \frac{1}{2}p_0(ct - x) + \frac{1}{2}p_0(ct + x)$$



Acoustic wave equation: external source

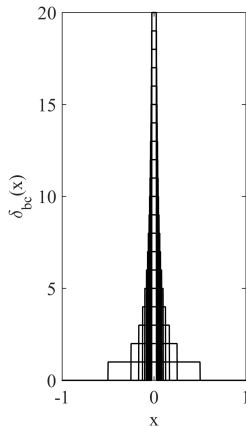
Green's Function G

$$\partial_t^2 G(\mathbf{x}, t; \mathbf{x}_0, t_0) - c^2 \Delta G(\mathbf{x}, t; \mathbf{x}_0, t_0) = \delta(\mathbf{x} - \mathbf{x}_0) \delta(t - t_0)$$

Delta function δ

$$\delta(x) = \begin{cases} \infty & x = 0 \\ 0 & x \neq 0 \end{cases}$$

$$\int_{-\infty}^{\infty} \delta(x) dx = 1, \quad \int_{-\infty}^{\infty} f(x) \delta(x) dx = f(0)$$



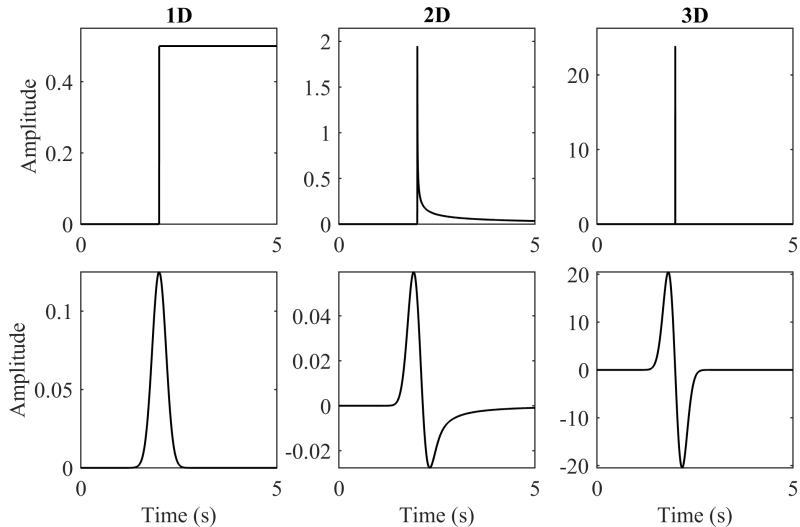
δ -generating function using
boxcars.

Acoustic wave equation: analytical solutions

Green's functions for the inhomogeneous acoustic wave equation for all dimensions. $H(t)$ is the Heaviside function.

1D	2D	3D
$\frac{1}{2c} H\left(t - \frac{ r }{c}\right)$	$\frac{1}{2\pi c^2} \frac{H\left(t - \frac{ r }{c}\right)}{\sqrt{t^2 - \frac{r^2}{c^2}}}$	$\frac{1}{4\pi c^2 r} \delta(t - r/c)$
$r = x$	$r = \sqrt{x^2 + y^2}$	$r = \sqrt{x^2 + y^2 + z^2}$

Acoustic wave equation: analytical solutions



The Elastic Wave Equation

Displacement-stress formulation

$$\rho \partial_t^2 u_i = \partial_j (\sigma_{ij} + M_{ij}) + f_i$$

$$\sigma_{ij} = \lambda \epsilon_{kk} \delta_{ij} + 2 \mu \epsilon_{ij}$$

$$\epsilon_{kl} = \frac{1}{2} (\partial_k u_l + \partial_l u_k),$$

Dependencies

u_i	$\rightarrow u_i(\mathbf{x}, t)$	$i = 1, 2, 3$
v_i	$\rightarrow v_i(\mathbf{x}, t)$	$i = 1, 2, 3$
σ_{ij}	$\rightarrow \sigma_{ij}(\mathbf{x}, t)$	$i, j = 1, 2, 3$
ϵ_{ij}	$\rightarrow \epsilon_{ij}(\mathbf{x}, t)$	$i, j = 1, 2, 3$
ρ	$\rightarrow \rho(\mathbf{x})$	
c_{ijkl}	$\rightarrow c_{ijkl}(\mathbf{x})$	$i, j, k, l = 1, 2, 3$
f_i	$\rightarrow f_i(\mathbf{x}, t)$	$i = 1, 2, 3$
M_{ij}	$\rightarrow M_{ij}(\mathbf{x}, t)$	$i, j = 1, 2, 3,$

1-D elastic wave equation

Shear Motion

$$\rho(x)\partial_t^2 u(x, t) = \partial_x [\mu(x)\partial_x u(x, t)] + f(x, t)$$

u displacement

f external force

ρ mass density

μ shear modulus

Velocity - Stress Formulation

Defining velocity v and stress component σ as

$$\partial_t u = v$$

$$\sigma = \mu \partial_x u$$

and assuming space-time dependencies leads to the wave equation

$$\rho \partial_t v = \partial_x \sigma + f$$

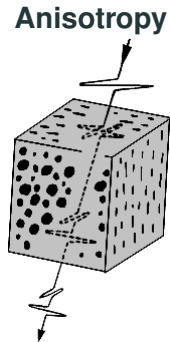
$$\dot{\sigma} = \mu \partial_x v$$

Our unknown solution vector is

$$\mathbf{q}(x, t) = (v, \sigma)$$

In order of relevance

- Viscoelasticity
- Anisotropy
- Poroelasticity
- Plasticity

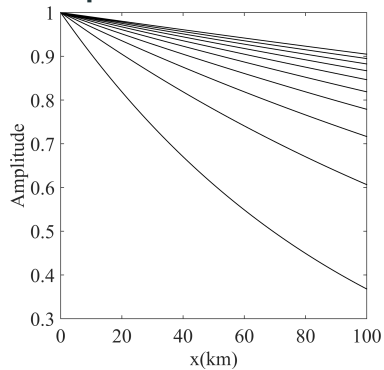


Amplitude decay

$$\frac{1}{Q(\omega)} = -\frac{\Delta E}{2\pi E}$$

$$A(x) = A_0 e^{-\frac{\omega x}{2cQ}}$$

Examples



Generalized Hooke's Law

$$\sigma_{ij} = c_{ijkl} \epsilon_{kl}, \quad i, j, k, l = 1, 2, 3$$

Reduced notation (Kelvin-Voigt)

$$c_{pq} = \begin{pmatrix} c_{11} & c_{12} & c_{13} & 0 & 0 & 0 \\ c_{12} & c_{11} & c_{13} & 0 & 0 & 0 \\ c_{13} & c_{13} & c_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & c_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & c_{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{c_{11} - c_{12}}{2} \end{pmatrix}$$

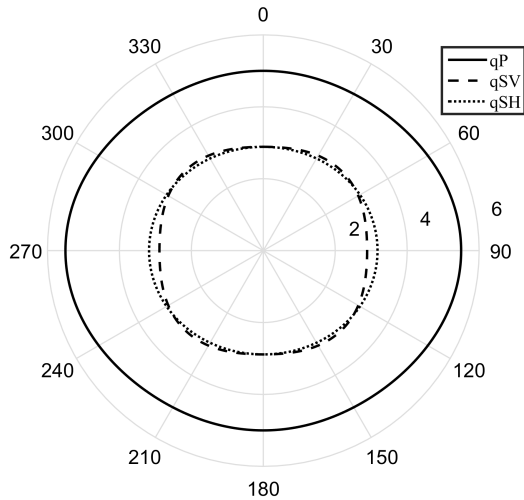
Velocity variations (Thomson parameters)

$$v_{qP}(\theta) = v_{P0} \left(1 + \delta \sin^2(\theta) \cos^2(\theta) + \epsilon \sin^4(\theta) \right)$$

$$v_{qSV}(\theta) = v_{S0} \left(1 + \frac{v_{P0}^2}{v_{S0}^2} (\epsilon - \delta) \sin^2(\theta) \cos^2(\theta) \right)_{(1)}$$

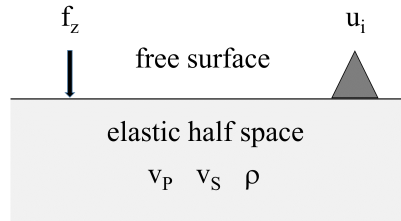
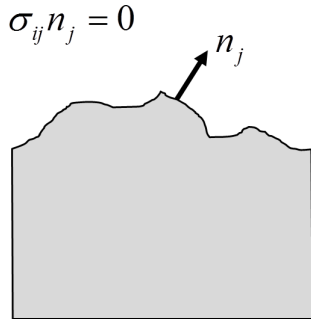
$$v_{qSH}(\theta) = v_{S0} \left(1 + \gamma \sin^2(\theta) \right)$$

Anisotropic velocities

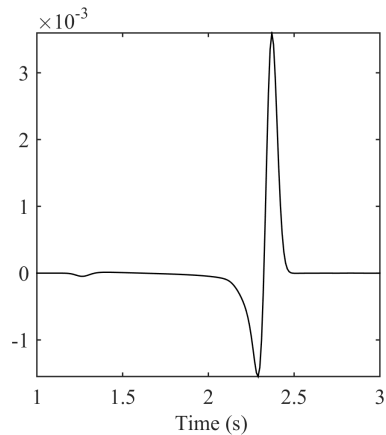
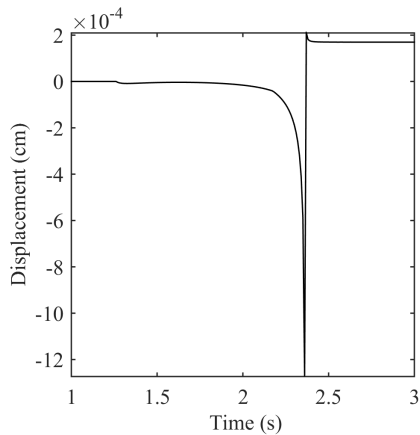


Free Surface Boundary Conditions

$$t_i = \sigma_{ij} n_j \rightarrow \sigma_{xz} = \sigma_{yz} = \sigma_{zz} = 0$$



Lamb's Problem (analytical solution)

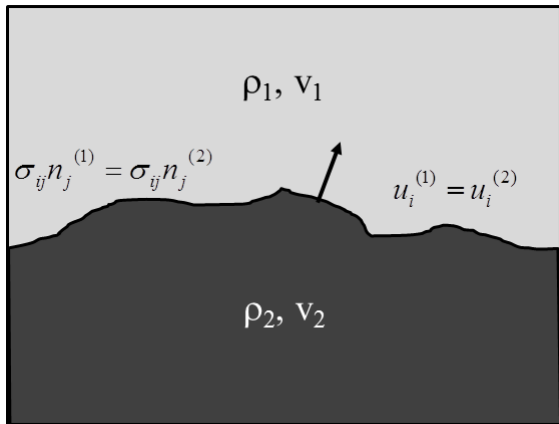


Internal Boundary Conditions

$$\sigma_{ij} n_j^{(1)} = \sigma_{ij} n_j^{(2)}$$

$$u_i^{(1)} = u_i^{(2)}$$

Internal boundary conditions need not be modelled explicitly!



Gradient, Divergence, Curl

Gradient

$$\nabla \mathbf{u}(\mathbf{x}, t) = \partial_j u_i(\mathbf{x}, t) .$$

Deformation

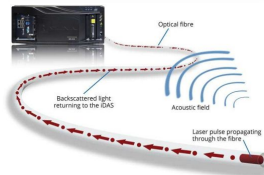
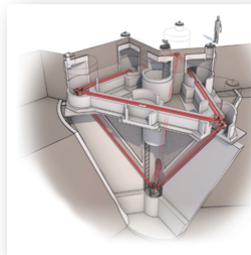
$$\epsilon_{ij}(\mathbf{x}, t) = \frac{1}{2}(\partial_i u_j(\mathbf{x}, t) + \partial_j u_i(\mathbf{x}, t))$$

Curl

$$\frac{1}{2} \nabla \times \mathbf{u} = \frac{1}{2} \begin{pmatrix} \partial_y u_z - \partial_z u_y \\ \partial_z u_x - \partial_x u_z \\ \partial_x u_y - \partial_y u_x \end{pmatrix}$$

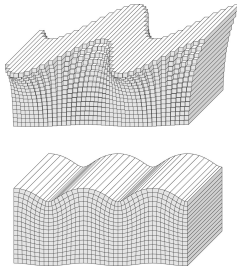
Divergence

$$\nabla \cdot \mathbf{u} = \partial_x u_x + \partial_y u_y + \partial_z u_z$$

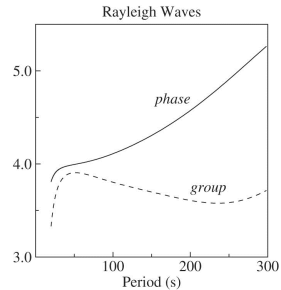
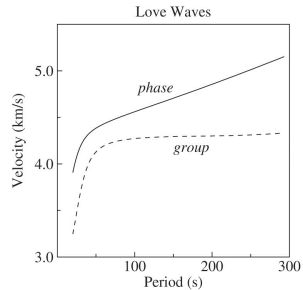


Surface Waves - Dispersion

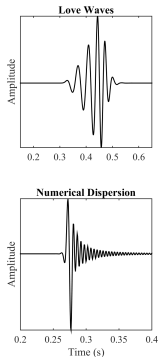
Ralyeigh and Love Waves



Dispersion Curves



Numerical Dispersion



Numerical and physical dispersion can
be confused!

In wave simulations we have to **avoid**
numerical dispersion!

Seismic Point Source - The Moment Tensor

Moment Tensor

$$\mathbf{M} = \begin{pmatrix} M_{11} & M_{12} & M_{13} \\ M_{21} & M_{22} & M_{23} \\ M_{31} & M_{32} & M_{33} \end{pmatrix}$$

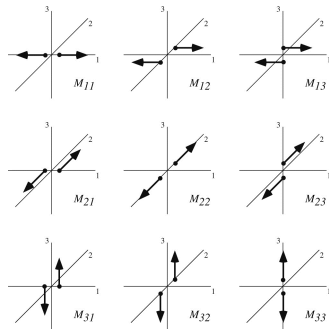
$$M_0 = \mu A d$$

M_0 scalar moment

μ shear modulus

A fault Area

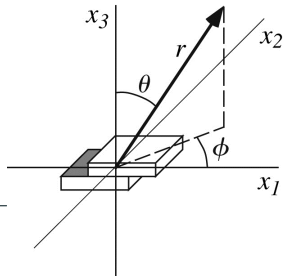
d average displacement



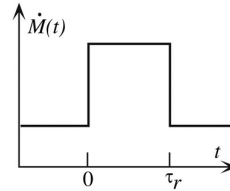
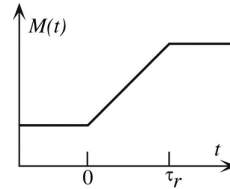
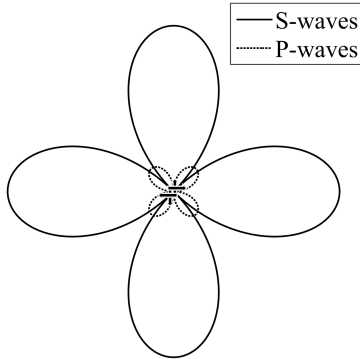
The DC analytical solution

Double Couple Green's function

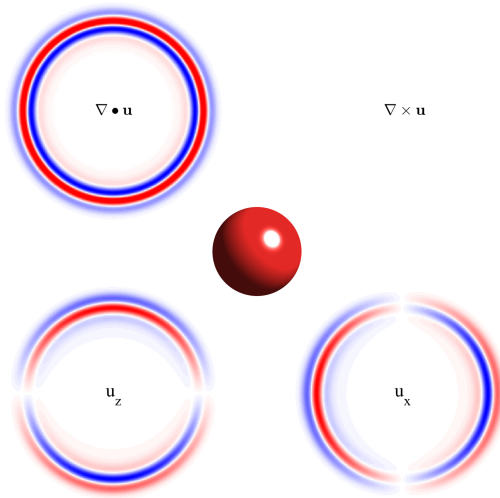
$$\begin{aligned} u(\mathbf{x}, t) = & \frac{1}{4\pi\rho} \mathbf{A}^N \frac{1}{r^4} \int_{r/\alpha}^{r/\beta} \tau M_0(t - \tau) d\tau + \\ & + \frac{1}{4\pi\rho\alpha^2} \mathbf{A}^{IP} \frac{1}{r^2} M_0(t - \frac{r}{\alpha}) + \frac{1}{4\pi\rho\beta^2} \mathbf{A}^{IS} \frac{1}{r^2} M_0(t - \frac{r}{\beta}) + \\ & + \frac{1}{4\pi\rho\alpha^3} \mathbf{A}^{FP} \frac{1}{r} \dot{M}_0(t - \frac{r}{\alpha}) + \frac{1}{4\pi\rho\beta^3} \mathbf{A}^{FS} \frac{1}{r} \dot{M}_0(t - \frac{r}{\beta}) \end{aligned}$$



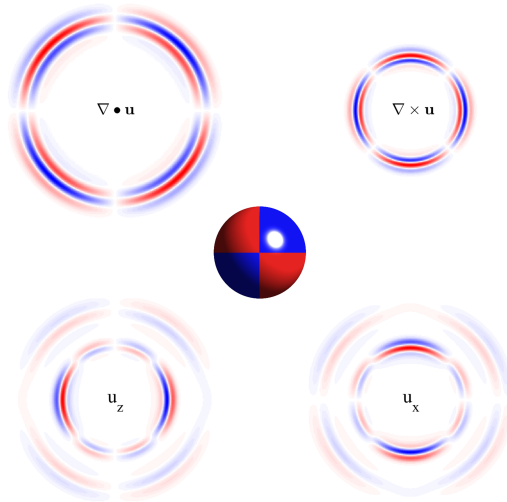
Radiation and Source Time Function



Wavefields from Moment Tensor Sources



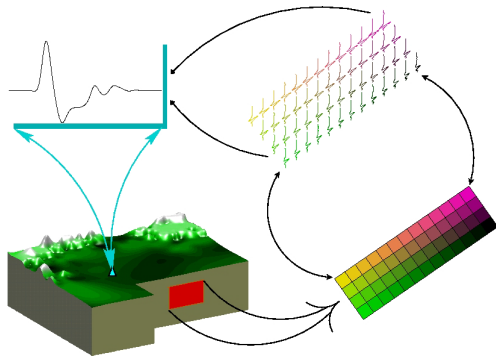
Wavefields from Moment Tensor Sources



Superposition Principle

$$v_j^r(\omega) = \sum_{k=1}^N \text{slip}_k \exp[-i\omega t_k(c^{rup})] G_{kl}^r(\omega) S(R, \omega)$$

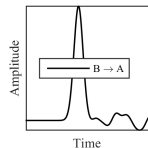
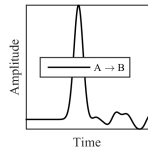
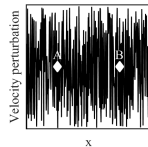
Finite sources can be simulated by summing up over point sources



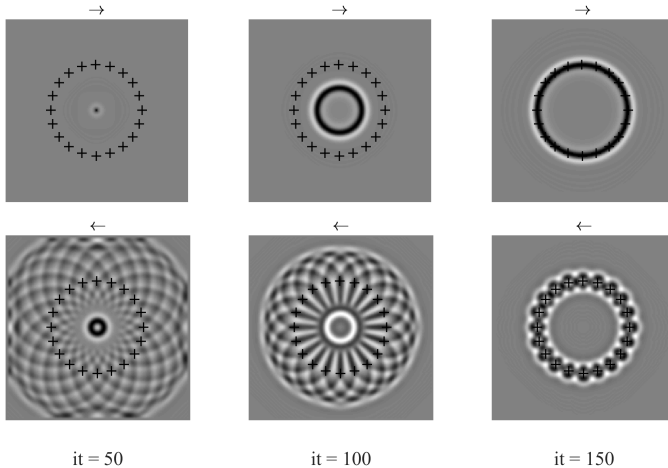
Reciprocity

$$G_{ij}(\mathbf{x}, t; \mathbf{x}_0, t_0) = G_{ji}(\mathbf{x}_0, -t_0; \mathbf{x}, -t)$$

The wave equation is symmetric in time. Source and receiver locations can be interchanged. This has dramatic consequences for modeling and inversion!



Time Reversal



Wave Equation as Linear System

Seismogram for arbitrary source $s(t)$ as convolution (exact)

$$p(\mathbf{x}, t) = G(\mathbf{x}, t, \mathbf{x}_0) \otimes s(t)$$

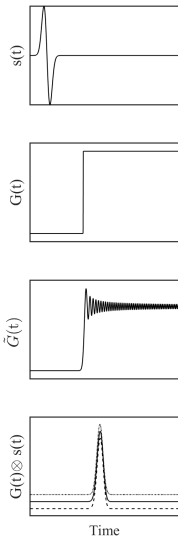
Seismogram for arbitrary source $s(t)$ as convolution (numerical)

$$\tilde{p}(\mathbf{x}, t) = \tilde{G}(\mathbf{x}, t, \mathbf{x}_0) \otimes s(t)$$

Important consequence:

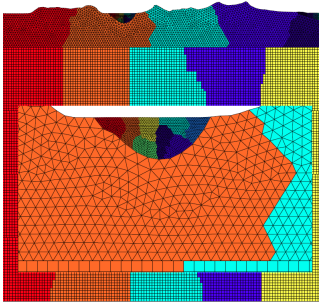
Even if your numerical Green's function $\tilde{G}(\mathbf{x}, t, \mathbf{x}_0)$ is inaccurate, the numerical solution $\tilde{p}(\mathbf{x}, t)$ might be very accurate provided the $s(t)$ is defined in the right frequency band!

Wave Equation as Linear System



- Exact Green's functions cannot be calculated numerically (infinite frequencies!)
- A numerical solver is a **linear system**
- The convolution theorem applies
- Inaccurate simulations can be filtered afterwards
- Source time functions can be altered afterwards
- ... provided the sampling is good enough ...

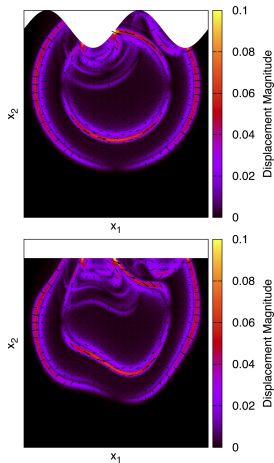
Challenges - Meshing



Human time	Simulation workflow	cpu time
15%	Design	0%
80% (weeks)	Geometry creation, meshing	10%
5%	Solver	90%

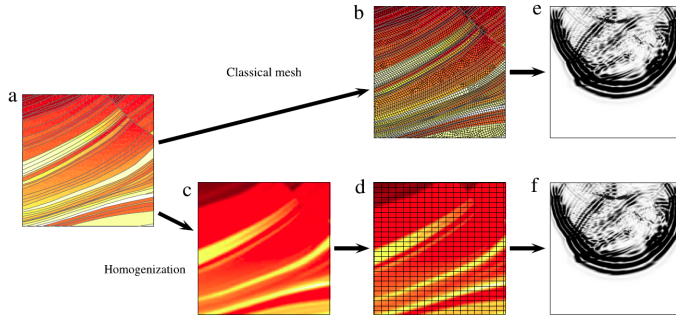
- Meshing work flow not well defined
- Still major bottleneck for simulation tasks with complex geometries
- Tetrahedral meshes easier, but ...
- Salvus?

Future Strategies - Alternative Formulations



- Particle relabelling, grid stretching
- Mapping geometrical complexity onto regular grids
- Smart pre-processing rather than meshing?
- Similar concept used in summation-by-parts (SBP) algorithms (SW4)

Future Strategies - Homogenization



- We only see low-pass filtered Earth
- So why simulate models with infinite frequencies?
- Homogenisation of discontinuous model
- Renaissance of regular grid methods?

- What physical parameters make up an **isotropic (anisotropic, viscoelastic)** Earth model?
- What is the definition of **dispersion** in simple words?
- What are **analytical** solutions to partial differential equations good for?
- Which parameters describe a **seismic source** in a simulation?
- What seismic **wave types** do you expect for a simulation with an **explosion source** in a homogeneous medium?

- What physical parameters make up an **isotropic (anisotropic, viscoelastic)** Earth model?
- What is the definition of **dispersion** in simple words?
- What are **analytical** solutions to partial differential equations good for?
- Which parameters describe a **seismic source** in a simulation?
- What seismic **wave types** do you expect for a simulation with an **explosion source** in a homogeneous medium?

- What physical parameters make up an **isotropic (anisotropic, viscoelastic)** Earth model?
- What is the definition of **dispersion** in simple words?
- What are **analytical** solutions to partial differential equations good for?
- Which parameters describe a **seismic source** in a simulation?
- What seismic **wave types** do you expect for a simulation with an **explosion source** in a homogeneous medium?

- What physical parameters make up an **isotropic (anisotropic, viscoelastic)** Earth model?
- What is the definition of **dispersion** in simple words?
- What are **analytical** solutions to partial differential equations good for?
- Which parameters describe a **seismic source** in a simulation?
- What seismic **wave types** do you expect for a simulation with an **explosion source** in a homogeneous medium?

- What physical parameters make up an **isotropic (anisotropic, viscoelastic)** Earth model?
- What is the definition of **dispersion** in simple words?
- What are **analytical** solutions to partial differential equations good for?
- Which parameters describe a **seismic source** in a simulation?
- What seismic **wave types** do you expect for a simulation with an **explosion source** in a homogeneous medium?

Questions?